# Acceleration caused by gravity



AS Level | **9702 AS Ch.2** | Chapter 2: Accelerated motion | Lesson 2.11 | Sections 2.11

**Goal:** Use g = 9.81 m s⁻² in the suvat equations for objects falling, or thrown up, without air resistance.

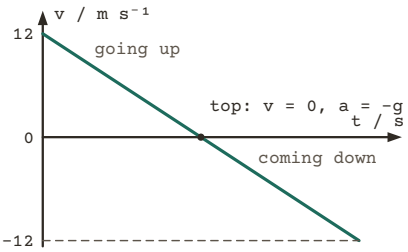
## 01 The big picture

Near the Earth's surface, with no air resistance, every object falls with the same acceleration $g = 9.81$ m s⁻², downwards, whatever its mass. This is free fall.

An object thrown upwards has the same downward acceleration all the way: it slows, stops for an instant at the top (where $v = 0$ but $a$ is still $g$) and speeds up on the way down. Take up as positive and use $a = -9.81$ m s⁻², or down as positive and $a = +9.81$.

**Acceleration of free fall $g$** [paraphrase]
The acceleration of a falling object with no air resistance: 9.81 m s⁻² near the Earth.

**Fig. 1 · v-t for a ball thrown up**




One straight line with gradient $-g$ all the way, through $v = 0$ at the top.

## 02 Key equations

**Dropped from rest: time**

$$s = \tfrac{1}{2}gt^2$$

**Dropped from rest: speed**

$$v^2 = 2gs$$

**Symbols and units**

| | | | | | |
|---|---|---|---|---|---|
| $g$ | acceleration of free fall, 9.81 | $\mathrm{m\,s^{-2}}$ | $s$ | distance fallen | $\mathrm{m}$ |

## 03 Sample problems with solutions

**Sample 1 · A dropped stone**

A stone is dropped from rest from a bridge 20 m above the water. Ignoring air resistance, calculate the time it takes to fall and its speed when it hits the water.

**Step 1.** Down positive, $u = 0$, $a = 9.81$:

$$t = \sqrt{\frac{2s}{g}} = \sqrt{\frac{2 \times 20}{9.81}} = 2.02\ \mathrm{s}$$

**Step 2.** Speed:

$$v = \sqrt{2gs} = \sqrt{2 \times 9.81 \times 20} = 19.8\ \mathrm{m\,s^{-1}}$$

**Answer:** 2.0 s; 20 m s⁻¹ (19.8 m s⁻¹)

Unit check: $\sqrt{\mathrm{m\,s^{-2}} \times \mathrm{m}} = \mathrm{m\,s^{-1}}$ ✓

## 04 Practice problems

**P1** A ball is thrown up at 12 m s⁻¹. How long does it take to reach the top?
Answer: 1.2 s (1.22 s)

**P2** How high does that ball rise?
Answer: 7.3 m

**P3** A ball is dropped from rest. Find its speed and the distance fallen after 3.0 s.
Answer: 29 m s⁻¹ (29.4); 44 m (44.1)

**P4** A hammer and a feather are dropped together on the Moon (no air). Which lands first?
Answer: Neither: they land together

## 05 Other useful information

**Common mistakes**

✗ **Wrong:** At the top of its flight the acceleration is zero.
✓ **Right:** Only $v$ is zero; $a$ is still 9.81 m s⁻² downwards.

**Exam tips**

- Write your sign convention ("up +") at the top of the working.

**What the syllabus asks**

- 9702 topic 2: free fall without air resistance. Coursebook 2.11.

**Explore:** Practice questions for this lesson